PROJECT PROGRESS REPORT

Project Title: IoT-Based Smart Soil Health Monitoring and Crop Recommendation System

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1. Project Planning & Task Definition

1. Project Planning

The development of the *IoT-Based Smart Soil Health Monitoring and Crop Recommendation System* was structured into clearly defined phases to ensure systematic progress, effective coordination, and timely completion of milestones. A phased development model was adopted to minimize risk and allow iterative improvements.

Phase 1: Literature Review and Problem Analysis

- Studied existing IoT-based smart agriculture systems.
- Analyzed research papers on soil health monitoring and crop prediction models.
- Identified limitations in current systems such as lack of real-time NPK monitoring and absence of integrated fertilizer recommendation.
- Finalized the problem statement and defined system scope.

Phase 2: Requirement Specification and System Design

- Identified hardware and software requirements.
- Selected appropriate sensors (NPK, soil moisture, temperature, humidity).
- Designed the overall system architecture including data flow from sensors to cloud platform.
- Prepared block diagram and initial workflow design.

Phase 3: Hardware Setup and Sensor Integration

- Interfaced soil moisture sensor and DHT sensor with ESP32/ESP8266 microcontroller.

- Integrated NPK sensor and performed calibration tests.
- Established Wi-Fi connectivity for data transmission.
- Conducted initial hardware testing and debugging.

Phase 4: Data Collection and Cloud Integration

- Configured cloud platform (ThingSpeak/Firebase) for real-time data storage.
- Developed firmware to transmit sensor readings periodically.
- Verified successful real-time data upload and visualization.
- Created dataset for crop nutrient requirement mapping.

Phase 5: Crop Recommendation Logic Implementation

- Developed nutrient classification model based on NPK thresholds.
- Implemented rule-based matching algorithm for crop recommendation.
- Designed fertilizer suggestion logic based on nutrient deficiency levels.
- Tested recommendation logic using sample soil datasets.

Phase 6: Testing and Performance Evaluation

- Conducted lab-based testing of complete system.
- Verified accuracy of sensor readings.
- Evaluated response time and cloud data consistency.
- Compared recommendation results with standard agricultural guidelines.
- Identified areas for improvement and optimization.

1.1 Task Distribution

To ensure equal participation and effective execution, responsibilities were distributed based on individual strengths and technical expertise.

Member	Assigned Responsibilities
Aryan Saxena	Overall project coordination, IoT system integration, microcontroller programming, Wi-Fi configuration, cloud connectivity setup
Harshwardhan Pandey	Sensor interfacing (NPK, soil moisture, temperature, humidity), hardware assembly, calibration, circuit testing
Vinayak Sahu	Data preprocessing, soil nutrient analysis, crop recommendation algorithm development, fertilizer optimization logic
Sumit Kant	Documentation preparation, IEEE research paper drafting, literature review compilation, result analysis and reporting

2. Problem Statement

- Agriculture remains one of the most critical sectors for economic growth, food security, and rural livelihood development. However, traditional farming practices still rely largely on manual soil testing, seasonal laboratory analysis, and experience-based decision-making. These conventional methods are often time-consuming, costly, and inaccessible to small and marginal farmers. As a result, farmers frequently apply fertilizers without accurate knowledge of soil nutrient composition, leading to excessive or insufficient fertilizer usage.
- Improper nutrient management not only reduces crop productivity but also contributes to soil degradation, groundwater contamination, and environmental imbalance. Additionally, crop selection is often based on regional trends rather than real-time soil conditions, which may result in suboptimal yields and financial losses. Irrigation practices also lack scientific precision, causing either water wastage or crop stress due to insufficient watering.
- With increasing global population and climate variability, there is an urgent need for smart agricultural solutions that enable data-driven decision-making. Real-time soil health monitoring combined with intelligent crop recommendation systems can significantly improve productivity while optimizing resource utilization.
- Therefore, this project aims to develop a cost-effective, IoT-based smart soil health monitoring system capable of measuring essential soil parameters and generating actionable crop, fertilizer, and irrigation recommendations. The system is designed to assist farmers in making informed decisions, thereby improving agricultural efficiency and sustainability.

2.1 Objectives

- The primary objective of this project is to design and implement an intelligent soil monitoring and crop recommendation system using Internet of Things (IoT) technology.
- The specific objectives are:
- • To design and develop a real-time IoT-based soil monitoring system capable of continuous data acquisition.
- • To measure critical soil parameters including Nitrogen (N), Phosphorus (P), Potassium (K), soil moisture, temperature, and humidity using appropriate sensors.
- • To establish reliable wireless communication between the sensor unit and cloud platform for real-time data transmission.
- • To analyze collected soil data to determine soil fertility status and nutrient deficiency levels.
- • To develop a crop recommendation model that matches soil conditions with suitable crop requirements.
- • To implement fertilizer suggestion logic to optimize nutrient management and reduce excessive chemical usage.
- • To provide irrigation recommendations based on soil moisture and environmental conditions.
- • To promote precision farming practices that improve productivity while conserving resources.
- • To contribute toward sustainable agriculture and align with global Sustainable Development Goals (SDGs).

3. Requirement Specification & System Architecture

The proposed IoT-Based Smart Soil Health Monitoring and Crop Recommendation System requires a combination of hardware and software components to ensure accurate data acquisition, reliable communication, and intelligent data processing. The requirement specification defines the essential resources needed for successful implementation.

3.1 Hardware Requirements

The hardware components form the core sensing and communication layer of the system. Each component is selected based on cost-effectiveness, reliability, and compatibility.

• NPK Soil Sensor

The NPK sensor is used to measure the concentration of essential soil nutrients — Nitrogen (N), Phosphorus (P), and Potassium (K). These nutrients play a critical role in plant growth and crop productivity. The sensor provides real-time nutrient values, enabling accurate soil fertility assessment and fertilizer recommendation.

• Soil Moisture Sensor

The soil moisture sensor measures the volumetric water content in soil. It helps determine irrigation requirements and prevents overwatering or underwatering. This improves water management and supports precision irrigation.

• DHT11 / DHT22 Sensor

The DHT sensor is used to measure environmental temperature and humidity. These parameters influence crop growth and soil conditions. DHT22 is preferred for higher accuracy and wider operating range.

• ESP8266 / ESP32 Microcontroller

The ESP8266 or ESP32 acts as the central processing and communication unit. It collects sensor data, processes basic logic, and transmits data to the cloud via Wi-Fi. ESP32 is preferred due to:

- Built-in Wi-Fi and Bluetooth
- Higher processing capability
- Multiple GPIO pins
- Low power consumption

• Power Supply Module

A regulated power supply ensures stable voltage for sensors and microcontroller. This may include:

- Rechargeable battery pack
- Voltage regulator module
- Optional solar panel for sustainable deployment

3.2 Software Requirements

The software layer enables programming, data storage, analysis, and recommendation generation.

• Arduino IDE

Used for writing and uploading firmware to ESP32/ESP8266. It handles:

- Sensor data acquisition
- Wi-Fi connectivity setup
- Data transmission protocols

• Python Programming Environment

Python is used for backend data processing and recommendation logic implementation. It supports:

- Data preprocessing
- Soil fertility classification
- Crop matching algorithms
- Fertilizer suggestion model

• Cloud Platform (ThingSpeak / Firebase)

Cloud platforms are used for:

- Real-time data storage
- Remote monitoring
- Data visualization
- Historical data analysis

ThingSpeak is suitable for quick IoT dashboards, while Firebase supports scalable real-time database solutions.

• Data Analysis Libraries

Python libraries such as:

- NumPy
- Pandas
- Matplotlib
- Scikit-learn (for future ML extension)

These libraries assist in data analysis, visualization, and model development.

3.3 System Architecture

The system architecture follows a layered IoT framework consisting of sensing, communication, cloud, and decision-making layers.

Data Flow Model

Sensors → Microcontroller → Cloud Database → Data Processing Engine → Crop Recommendation Output

Layered Architecture Description

1. Sensing Layer

This layer includes NPK, soil moisture, and DHT sensors. It is responsible for real-time collection of soil and environmental data.

2. Processing & Communication Layer

The ESP32 microcontroller collects sensor readings and converts them into digital values. It establishes Wi-Fi connectivity and transmits data securely to the cloud server at regular

intervals.

3. Cloud Storage Layer

The cloud platform stores real-time and historical data. It enables remote access, monitoring, and dashboard visualization.

4. Data Processing & Analytics Layer

Python-based algorithms analyze:

- Nutrient concentration levels
- Soil fertility classification (Low, Medium, High)
- Crop requirement matching
- Deficiency detection

5. Recommendation Layer

Based on analytical results, the system generates:

- Suitable crop suggestions
- Fertilizer quantity recommendations
- Irrigation guidance

3.4 Working Principle

1. Sensors continuously monitor soil and environmental parameters.
2. ESP32 reads sensor values at predefined intervals.
3. Data is transmitted to the cloud via Wi-Fi.
4. Cloud database stores and processes the data.

5. Analytical model compares soil parameters with crop requirement dataset.

6. System outputs recommendations through dashboard or mobile interface.

4. Literature Review / Background Study

The rapid advancement of Internet of Things (IoT) technology has significantly transformed modern agricultural practices. Numerous research studies have explored the application of IoT in smart farming to enhance productivity, optimize resource utilization, and enable data-driven decision-making. Researchers have primarily focused on real-time environmental monitoring, automated irrigation systems, and wireless sensor networks for soil moisture detection.

Several existing smart agriculture systems integrate soil moisture sensors with microcontrollers such as Arduino or ESP8266 to automate irrigation processes. These systems successfully reduce water wastage and improve irrigation efficiency. However, most of these solutions are limited to monitoring only moisture levels and temperature conditions, without incorporating detailed soil nutrient analysis.

Some research works have proposed cloud-based agricultural monitoring systems where sensor data is uploaded to platforms like ThingSpeak or Firebase for remote visualization. While these systems provide real-time monitoring and remote access, they often lack intelligent decision-support mechanisms such as crop suitability analysis and fertilizer optimization.

Recent studies have introduced machine learning approaches for crop prediction based on historical datasets. However, many of these models rely on static or previously collected soil datasets rather than real-time soil nutrient values. The absence of live NPK monitoring reduces the practical applicability of such systems in dynamic field conditions.

A major gap identified in the literature is the lack of integrated systems that combine:

- Real-time NPK soil nutrient monitoring
- Soil moisture and environmental parameter sensing
- Cloud-based data analytics
- Crop recommendation models
- Fertilizer and irrigation advisory systems

Most existing solutions address only one or two aspects of smart agriculture rather than providing a comprehensive soil health management framework.

Limitations in Existing Systems

Based on the literature review, the following limitations were observed:

- Focus primarily on irrigation control rather than complete soil fertility analysis
- Limited integration of real-time NPK monitoring
- Lack of actionable fertilizer recommendation mechanisms
- Minimal decision-support systems for crop selection
- Dependence on manual interpretation of data

Contribution of the Proposed System

The proposed IoT-Based Smart Soil Health Monitoring and Crop Recommendation System addresses these research gaps by offering a comprehensive and integrated solution.

The key improvements over existing systems include:

- Integration of multiple soil parameters including NPK, moisture, temperature, and humidity
- Real-time cloud-based monitoring with continuous data logging
- Development of a crop recommendation model based on soil nutrient thresholds
- Implementation of fertilizer optimization logic to reduce excessive chemical usage
- Support for precision farming practices
- Alignment with sustainable agriculture goals and environmental conservation

By combining sensing, cloud computing, and intelligent analytics into a unified system, the proposed model enhances decision-making accuracy and provides practical value to farmers.

5. Methodology & Technical Approach

The proposed IoT-Based Smart Soil Health Monitoring and Crop Recommendation System follows a structured and systematic methodology to ensure accurate data acquisition, reliable processing, and intelligent recommendation generation. The methodology is divided into four major stages: Data Collection, Data Transmission, Data Processing, and Recommendation Generation.

5.1 Step 1: Data Collection

The first stage of the system involves real-time acquisition of soil and environmental parameters using IoT sensors. Accurate data collection is essential for effective soil fertility assessment and crop recommendation.

The following parameters are measured:

- **Nitrogen (N)**

Nitrogen is essential for leaf growth and overall plant development. The NPK sensor measures nitrogen concentration in the soil, which helps determine nutrient deficiency or excess.

- **Phosphorus (P)**

Phosphorus supports root development and flowering. Monitoring phosphorus levels helps in evaluating soil fertility and crop suitability.

- **Potassium (K)**

Potassium enhances disease resistance and improves crop quality. Real-time potassium measurement supports balanced fertilizer recommendations.

- **Soil Moisture**

The soil moisture sensor measures volumetric water content in soil. This parameter is used to determine irrigation requirements and prevent overwatering or drought stress.

- **Temperature**

Temperature affects seed germination, nutrient absorption, and crop growth cycles. Environmental temperature is measured using a DHT sensor.

- **Humidity**

Humidity influences evapotranspiration rates and crop health. Monitoring humidity helps in analyzing environmental conditions for optimal farming.

All sensor data is collected at predefined intervals to ensure continuous monitoring.

5.2 Step 2: Data Transmission

After collecting sensor readings, the ESP32 microcontroller processes the raw data and prepares it for transmission.

The ESP32 performs the following functions:

• Converts analog sensor signals into digital values
• Formats data into structured packets
• Establishes Wi-Fi connectivity
• Sends data securely to the cloud platform

The data is transmitted to a cloud database (ThingSpeak/Firebase) using HTTP or MQTT protocols. This enables:

- Real-time remote monitoring
- Centralized data storage
- Historical data analysis
- Dashboard visualization

Reliable wireless communication ensures uninterrupted data flow between the field unit and cloud server.

5.3 Step 3: Data Processing & Analysis

Once uploaded to the cloud, the collected data undergoes preprocessing and analytical evaluation using Python-based algorithms.

Data Preprocessing

• Removal of noise and inconsistent readings
• Handling missing values
• Normalization of nutrient values
• Threshold classification (Low, Medium, High)

Nutrient Level Analysis

The system compares measured NPK values with standard agricultural threshold ranges to determine soil fertility status. Soil classification is performed based on nutrient availability levels.

Soil Classification

The soil is categorized into:

- Nutrient Deficient
- Moderately Fertile
- Highly Fertile

This classification helps in selecting suitable crops.

Crop Requirement Database Matching

A crop dataset containing nutrient requirements for various crops is maintained. The algorithm matches soil nutrient values with crop nutrient requirements using rule-based logic.

Example:
If N = Medium, P = High, K = Medium ⟶ Suitable crops: Wheat, Maize

Future improvements may include machine learning models to enhance prediction accuracy.

5.4 Step 4: Recommendation Generation

Based on analytical results, the system generates intelligent and actionable recommendations.

Crop Recommendation

The system suggests suitable crops that can grow efficiently under the detected soil

conditions. This helps farmers make informed crop selection decisions.

Fertilizer Recommendation

If nutrient deficiency is detected, the system calculates:

- Required nutrient supplementation
- Suggested fertilizer type
- Estimated fertilizer quantity

This prevents overuse of chemical fertilizers and promotes balanced nutrient management.

Irrigation Recommendation

Based on soil moisture and environmental parameters, the system determines:

- Whether irrigation is required
- Recommended water quantity
- Frequency of irrigation

This improves water efficiency and supports precision irrigation.

5.5 Overall Technical Workflow

1. Sensors continuously monitor soil and environmental conditions.
2. ESP32 collects and transmits data to cloud server.
3. Cloud platform stores real-time and historical data.
4. Python analytics engine processes soil parameters.
5. System generates crop, fertilizer, and irrigation recommendations.
6. Results are displayed through dashboard or user interface.

Advantages of the Proposed Methodology

✔ Real-time monitoring and decision support
✔ Data-driven crop selection
✔ Optimized fertilizer usage
✔ Efficient water management
✔ Reduced environmental impact
✔ Scalable and cost-effective design

6. Implementation Progress / Model Development

The implementation phase of the IoT-Based Smart Soil Health Monitoring and Crop Recommendation System has progressed through structured hardware integration, cloud configuration, and model development stages. Significant milestones have been achieved in both hardware setup and software logic development.

6.1 Work Completed So Far

• Sensor Interfacing with ESP32

The soil moisture sensor and DHT (temperature and humidity) sensor have been successfully interfaced with the ESP32 microcontroller. Proper GPIO configuration and analog-to-digital conversion were implemented to ensure accurate sensor readings.

The following tasks were completed:

- GPIO pin configuration
- Calibration of soil moisture sensor
- Testing of DHT sensor accuracy
- Serial monitoring of real-time values

- Error handling for unstable readings

Initial hardware debugging was conducted to eliminate noise and fluctuation in sensor outputs.

• Real-Time Moisture and Temperature Monitoring

The system is currently capable of continuously monitoring:

- Soil moisture levels
- Environmental temperature
- Relative humidity

Sensor data is updated at predefined time intervals, ensuring near real-time observation. The monitoring system was tested under controlled lab conditions to verify response time and measurement consistency.

• Basic Cloud Data Upload

Wi-Fi connectivity using ESP32 has been successfully established. Sensor readings are transmitted to the cloud platform using HTTP protocol.

Achievements include:

- Wi-Fi authentication setup
- API integration with cloud platform
- Real-time data logging
- Visualization dashboard configuration
- Historical data storage testing

Cloud data can now be accessed remotely for monitoring and analysis.

• Crop Dataset Preparation

A structured dataset containing crop nutrient requirements was developed. The dataset includes:

- Crop name
- Required Nitrogen range
- Required Phosphorus range
- Required Potassium range
- Ideal moisture level
- Suitable temperature range

This dataset forms the foundation of the crop recommendation engine.

• Initial Recommendation Logic Developed

A rule-based algorithm has been implemented in Python to match soil parameters with crop requirements.

The logic performs:

- Nutrient threshold comparison
- Soil fertility classification (Low/Medium/High)
- Matching with suitable crop list
- Basic fertilizer deficiency detection

Preliminary testing of the algorithm using sample datasets has shown promising results.

6.2 Work Currently In Progress

• Full NPK Integration Testing

The integration of the NPK sensor is currently under testing and calibration. The following steps are being performed:

- Calibration against reference values
- Stability testing under varying soil conditions
- Data filtering to remove sensor noise
- Integration with cloud transmission module

Once stabilized, NPK values will be fully incorporated into the recommendation engine.

• Fertilizer Optimization Model

The fertilizer recommendation module is being enhanced to provide more precise suggestions. This includes:

- Deficiency gap calculation
- Estimation of required nutrient supplementation
- Mapping nutrient deficit to appropriate fertilizer types
- Reducing excessive chemical application

Future improvements may include predictive modeling using machine learning.

• Web Dashboard Development

A user-friendly dashboard is under development to display:

- Real-time soil parameter values

- Graphical data trends
- Crop recommendation results
- Fertilizer and irrigation suggestions

The dashboard will enable farmers or users to monitor soil health remotely and make informed decisions.

6.3 Technical Challenges Faced

During implementation, several practical challenges were encountered:

• Sensor value fluctuation due to soil inconsistency
• Calibration complexity of NPK sensor
• Wi-Fi instability during continuous transmission
• Data synchronization between cloud and processing module

These issues are being addressed through calibration refinement, error handling mechanisms, and improved data filtering techniques.

6.4 Current System Status

At present, the system prototype is functional with:

✔ Working sensor interface
✔ Real-time cloud connectivity
✔ Basic recommendation logic
✔ Structured crop dataset

The project is progressing toward full integration of NPK analysis and intelligent fertilizer optimization.

7. Results Achieved So Far

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7. Results Achieved So Far

The IoT-Based Smart Soil Health Monitoring and Crop Recommendation System has successfully completed initial prototype development and functional testing. The following results have been achieved during Phase-2 implementation.

7.1 Real-Time Soil Monitoring Results

The system is capable of collecting and displaying real-time soil and environmental data. The following parameters are currently monitored:

- Soil Moisture (%)
- Temperature (°C)
- Humidity (%)

Observed Performance:

• Stable sensor readings under controlled laboratory conditions
• Real-time data refresh at predefined intervals
• Consistent value transmission to cloud platform
• Minimal latency in data upload

The soil moisture sensor successfully detected changes when soil water content was varied, confirming proper functionality.

7.2 Cloud Connectivity and Data Logging

Cloud integration has been successfully implemented using Wi-Fi-enabled ESP32.

Achievements:

• Continuous data transmission to cloud database
• Real-time graphical visualization of parameters
• Historical data storage and retrieval
• Remote monitoring capability

Data logs confirm reliable communication between the microcontroller and cloud server.

7.3 Initial Crop Recommendation Accuracy

The rule-based recommendation algorithm was tested using sample soil parameter datasets.

Results:

• Correct classification of soil fertility levels (Low/Medium/High)
• Successful mapping of soil nutrient values to suitable crops
• Preliminary accuracy level estimated at approximately 75–80% under controlled dataset testing

The system correctly identified crop suitability based on nutrient thresholds.

7.4 Fertilizer Suggestion Output (Prototype Stage)

A basic fertilizer recommendation logic has been tested.

The system can:

• Detect nutrient deficiency
• Suggest nutrient supplementation
• Indicate whether nitrogen, phosphorus, or potassium is lacking

Full quantitative fertilizer estimation is currently under refinement.

7.5 System Reliability and Stability

During testing, the following observations were made:

✔ Stable ESP32 performance
✔ No major data packet loss
✔ Acceptable Wi-Fi connectivity stability
✔ Functional integration between sensing and cloud layers

Minor sensor fluctuations were observed, which are being addressed through calibration and data filtering techniques.

7.6 Limitations Observed

Although the prototype is functional, certain limitations remain:

• NPK sensor calibration is still in progress
• Recommendation model is currently rule-based (not ML-based)
• Field-level testing has not yet been conducted
• Dashboard interface is under development

These limitations will be addressed in the next phase.

7.7 Expected Improvements in Next Phase

The following enhancements are planned:

• Complete NPK integration
• Improved fertilizer optimization logic
• Machine learning-based crop prediction (future extension)
• Real-field testing and validation
• Fully functional web dashboard

- # 8. Conclusion & Future Scope
- # 8.1 Conclusion
- The proposed IoT-Based Smart Soil Health Monitoring and Crop Recommendation System demonstrates the practical implementation of precision agriculture using real-time

sensing and data analytics. The system successfully integrates soil and environmental sensors with a Wi-Fi-enabled microcontroller to collect and transmit real-time data to a cloud platform.

- The developed prototype is capable of monitoring key soil parameters such as moisture, temperature, humidity, and nutrient levels (NPK). Using a rule-based analytical model, the system classifies soil fertility and generates crop and fertilizer recommendations. Initial testing has confirmed stable sensor performance, reliable cloud connectivity, and functional recommendation logic.
- The project addresses critical challenges in traditional farming practices by:
- • Reducing dependency on manual soil testing
  • Minimizing excessive fertilizer usage
  • Supporting informed crop selection
  • Promoting efficient water management
  • Encouraging sustainable agricultural practices
- By combining IoT technology with intelligent decision support mechanisms, the system contributes toward data-driven farming and aligns with sustainable development goals related to food security, resource optimization, and climate resilience.
- The results achieved so far demonstrate the feasibility of deploying a cost-effective smart soil monitoring solution for small and medium-scale farmers.
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- ## 8.2 Future Scope
- Although the prototype has achieved significant milestones, several enhancements can be implemented to improve system performance and scalability.
- **• Machine Learning Integration**
- Future versions of the system can incorporate machine learning algorithms to improve crop prediction accuracy using large agricultural datasets. Predictive models can enhance decision-making under varying climatic conditions.
- **• Full-Scale NPK Calibration and Field Deployment**
- Extensive calibration of the NPK sensor under real field conditions will improve measurement accuracy. Field-level pilot testing will validate system reliability in practical agricultural environments.
- **• Mobile Application Development**
- A user-friendly mobile application can be developed to provide farmers with real-time notifications, recommendations, and historical data analysis.
- **• Advanced Fertilizer Optimization**
- The fertilizer module can be enhanced to calculate exact fertilizer quantities based on land area, crop type, and nutrient deficiency gap.
- **• Solar-Powered Deployment**
- Integration of solar energy systems can make the device self-sustainable and suitable for remote rural areas.
- **• Integration with Weather Forecast APIs**

- Incorporating weather forecasting data can further improve irrigation and crop planning decisions.
- **• Multi-Language Support**
- To increase accessibility, the system can support regional languages for farmer-friendly communication.
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- ## Final Statement
- The proposed system lays a strong foundation for smart agriculture solutions that combine IoT, cloud computing, and intelligent analytics. With further refinement and large-scale deployment, the system has the potential to significantly enhance agricultural productivity while ensuring environmental sustainability.